

A decision clock is not a trading clock

Daily learning note | 28 September 2026 | Q006a

Sidiq review status: Not yet reviewed.

Research question and hypothesis

Can a 09:00 New York research decision remain valid while the calendar correctly represents holidays, early closes and daylight-saving offsets? The hypothesis is that session-date eligibility, information availability and executable market time must be checked separately. A core-hours-only decision rule would reject every intended decision; a weekdays-only calendar would admit Thanksgiving.

Yesterday's Q005 validated source clocks. The weekly review then split Q006 into today's contract and fixture work and tomorrow's lookup/decision implementation. This preserves the existing provisional 09:00 policy while making its assumptions testable. Lawful market-data access is still unestablished; no prices, returns or alpha are measured today.

Beginner explanation and definitions

Think of a shop with a planning meeting before its doors open. The meeting can be scheduled for a real business day without taking place during customer hours. A research decision is similar: choosing a portfolio and executing a trade are different events. Yesterday's closing information may be useful at 09:00, but that does not mean today's 09:00 price is an available fill.

Session date: a local date with scheduled core trading. **Core open/close:** the planned regular market interval boundaries. **Early close:** a session ending sooner than usual. **Decision time:** when the research policy fixes its input set. **Previous session:** the latest listed session strictly before the decision's session date. **UTC offset:** local clock time minus UTC. **Fixture:** a small, explicit reference example used to test software. **Availability:** when information can actually be used.

Data, metrics and predeclared protocol

The bounded reference covers every date from 29 October through 30 November 2026. These are prospective scheduled calendar facts with synthetic decision cases, not observed trading records. The reference scope is NYSE cash-equity core trading; no claim is made for all venues in the wider planned universe.

Acceptance requires complete ordered dates, explicit closed days, correct early-close timing, expected UTC conversions, an independent November session-count check, six rejected corrupted copies and two identical replays. The 09:00 decision must precede each open by 30 minutes. Q006b will separately test actual timestamp rejection and lookup behavior. Today's diagnostic is not that public API.

Mathematics and worked example

For session s , define $D(s)$ as decision, $O(s)$ as core open and $C(s)$ as core close. Let $P(s)$ be the largest listed session date before s . The policy requires $D(s) < O(s) < C(s)$. Its separate core-time predicate is $O(s) \leq t < C(s)$: the open is included, the close excluded. This convention does not model auctions or promise execution. A daily bar labelled at $C(s)$ is still a legitimate summary of the completed session.

UTC instant = local wall time - UTC offset. October 30 uses offset -04:00, so 09:00 becomes 13:00 UTC. On November 2, offset -05:00 makes the same local decision 14:00 UTC. Holding UTC fixed would change the local policy. The fixture stores explicit expected instants, checked against the installed named-zone rules.

Date	Decision UTC	Core UTC interval	Previous session
30 October	13:00	13:30 to 20:00	29 October
2 November	14:00	14:30 to 21:00	30 October
27 November	14:00	14:30 to 18:00	25 November
30 November	14:00	14:30 to 21:00	27 November

Regular duration = (16 hours - 9.5 hours) * 60 = 390 minutes. Early-close duration = (13 hours - 9.5 hours) * 60 = 210 minutes. The 27 November session remains eligible for a 09:00 decision. Its predecessor is November 25, because November 26 is closed. Monday November 30 uses Friday November 27, whose completed bar ends at the early close. Neither a holiday nor a weekend needs a fabricated bar.

For a prior-session input, define $a = \max(\text{publication, receipt})$ and nonnegative processing delay l . It is available only if $a + l \leq D(s)$, in addition to session, validity and any staleness requirements. A synthetic prior-session value received at 08:59:30 with 30 seconds of processing meets 09:00 exactly. Received at 08:59:31 with the same delay, it is one second too late. Knowing a session has ended proves neither timely receipt nor that a later revision was already known. The calendar cannot replace Q001.

Primary sources and provenance

NYSE's 2026 schedule specifies November 26 closure and November 27's 13:00 equity close; the regular core interval is 09:30 to 16:00 Eastern. The separate estimated trading-day table gives November 20 sessions. <https://www.nyse.com/trade/hours-calendars>
https://www.nyse.com/publicdocs/Trading_Days.pdf

Python documents that zoneinfo uses external IANA rules. The consulted local database reports 2026c; its New York TZif hash is recorded separately from the calendar fixture's hash.
<https://docs.python.org/3/library/zoneinfo.html>

Sources were consulted on 28 September 2026. The exchange page exposes no release ID: the fixture names its consultation date and source edition without inventing a publisher version. It preserves a small transcription, not a downloaded market dataset or a complete archive of the source page.

Implementation and reproduction

The policy is `data/calendar-and-decisions.md`. The versioned reference is `experiments/calendar/nyse-2026-autumn-v1.json`. Every date has a status; session rows add expected UTC decision/open/close values and closed dates have no fake times. Source and timezone provenance are separate. A SHA-256 of the fixture bytes appears in `experiments/calendar/contract-audit.json`.

Run `make calendar-audit` to reproduce the diagnostic. `scripts/calendar_contract_audit.py` verifies all 33 records, their bounded statuses and UTC instants, cross-checks the November total and derives prior sessions and durations. Use `--output PATH` for scratch results. Run `make check` for the complete suite, `make manifest-experiment` for existing provenance checks and make note `NOTE_DATE=2026-09-28` for this PDF. No runtime dependency or package API changes. Metadata/schema generalization remains outside this audit.

Thirty-five new tests check source/policy metadata, a hand-listed weekend set, four independently worked UTC examples, all 22 session intervals, six corruptions and two complete subprocess replays. The corruptions remove a date, repeat a date, open the holiday, use the wrong post-change UTC hour, erase the early close or add an unexpected field. They fail the audit. These are fixture checks, not claims of production readiness.

Observed results and validation

Check	Observed result
Date coverage	33 dates; 22 sessions; 10 weekend dates; 1 holiday
Session types	21 regular; 1 early close
November source cross-check	20 sessions; weekday-only method incorrectly gives 21
Decision/open gap	30 minutes for all 22 sessions
Core durations	390 minutes regular; 210 minutes early close
Corrupted fixture copies	All 6 rejected
Replay	Two subprocess outputs match archived bytes
Python and quality	292 tests; Ruff lint/format; strict mypy on 16 source files pass
C++ and dependencies	Fresh warning-clean build and golden output; pip consistency pass
Prior source integrity	Q003's four pinned source files still verify

The first full check passed tests and lint but requested formatting of one generator expression. After formatting, the complete check passed. Restricted network fetch failed DNS; approved fetch succeeded. Main initially matched origin at 8228415. The authenticated issue read returned no open issues. The baseline weekly commit's CI passed; this does not establish CI success for today's eventual commit.

Limitations and interpretation

A planned schedule is not evidence that trading will actually occur. Emergency closures, security halts, auction mechanics, other venues, historical source vintages and the intended 2010-2025 data period remain outside this fixture. The hash identifies our bytes; it cannot authenticate the exchange or prove what was known at an earlier date. Timezone rules are not bundled.

Boundaries that the next implementation must preserve

Q006b must reject holidays, weekends, naive timestamps, wrong decision times and dates outside coverage. It must accept equivalent aware representations of 09:00, including valid pre-open decisions. Test exact boundaries and neighboring microseconds, the early close, UTC conversion and insufficient prior history. October 29 has no known predecessor; do not infer one outside the fixture. Twenty-session liquidity requires 20 actual prior sessions with usable observations, not 20 calendar days or repeated stale bars.

A source publication may validly arrive after market hours, on a holiday or over a weekend. Calendar policy must not reject such ingestion. The daily-feature policy restricts which completed session the value describes and whether it arrived by the decision. Eligibility loss cannot erase existing holdings, and a later execution still requires price, liquidity, halt and cost evidence.

For an interview, draw three points: yesterday's close, today's decision and today's open. Explain why session completion, information receipt and execution are distinct. Then move the UTC axis across the clock change and shorten Friday's session. State that these examples validate scheduling logic, not alpha.

Next work and necessary input

Q006a is complete; Q006b is ready for September 29. The Q006 parent remains incomplete until lookup and decision validation pass. Wednesday September 30 remains C002; corporate-action arithmetic follows. No input is needed for this independent engineering work. Actual source entitlement or source/budget input is still needed before Q004a acquisition, without sending credentials. Personal review remains Not yet reviewed and does not block publication or the next task.

Check your understanding

1. Why can a 09:00 decision be valid when the market opens at 09:30?
2. Why does a fixed 13:00 UTC decision fail to preserve the local policy after the autumn clock change?
3. Which prior session supplies November 27's daily inputs? Does that guarantee availability?
4. Does excluding the exact close from the core-time predicate invalidate a closing daily bar?
5. What should happen when a request needs a session outside fixture coverage?

Suggested answers

1. Decision validity depends on its local session date and the chosen research clock. Core-time validity is a separate predicate for scheduled market hours. A valid decision provides no fill guarantee.
2. New York's offset changes from -04:00 to -05:00. The same 09:00 local decision shifts from 13:00 UTC to 14:00 UTC; keeping 13:00 would instead select 08:00 local on the later dates.
3. November 25, because Thanksgiving is closed. The value must still be published, received, processed and valid by the decision; being a completed prior session is necessary but insufficient.
4. No. The half-open interval is a scheduling convention. A bar can summarize the session and be labelled at its close, with publication later. Auctions and execution at the close need their own model.
5. Fail explicitly for unknown coverage or insufficient history. Never assume a missing date is closed, generate an unverified weekday session or pad absent observations. Extend and verify the fixture first.